

Day 04 Assignment Submission

Question 1: Print Numbers from 1 to N

Program:

```
J PrintNum.java > ...  
1  import java.util.Scanner;  
2  
3  public class PrintNum{  
    Run | Debug  
4  public static void main(String[] args) {  
5      Scanner sc = new Scanner(System.in);  
6      System.out.print("Enter the value of n: ");  
7      int n = sc.nextInt();  
8      sc.close();  
9  
10     for ( int i = 1; i<=n ; i++)  
11         System.out.print(i + " ");  
12     }  
13 }
```

Output:

```
PS D:\Logic Building\Lab\Day 01> cd "d:\Logic  
Enter the value of n: 10  
1 2 3 4 5 6 7 8 9 10  
PS D:\Logic Building\Lab\Day 01> 
```

Question 2: Print Multiples of 3 between 1 and N

Program:

```
J MultiplesOfThree.java > ...  
1  import java.util.Scanner;  
2  
3  public class MultiplesOfThree {  
    Run | Debug  
4      public static void main(String[] args) {  
5          Scanner sc = new Scanner(System.in);  
6          System.out.print("Enter the value of n: ");  
7          int n = sc.nextInt();  
8          sc.close();  
9  
10         for ( int i = 1; i<= n ; i++){  
11             if (i % 3 == 0)  
12                 System.out.print(i + " ");  
13         }  
14     }  
15 }
```

Output:

```
PS D:\Logic Building\Lab\Day 01> cd "d:\Logic  
Enter the value of n: 20  
3 6 9 12 15 18  
PS D:\Logic Building\Lab\Day 01> 
```

Question 3: Calculate the Factorial of a Number

Program:

```
J Factorial.java > ...
1  import java.util.Scanner;
2
3  public class Factorial {
    Run | Debug
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6          System.out.print("Enter the value of n: ");
7          int n = sc.nextInt();
8          sc.close();
9
10         int fact = 1;
11
12         if (n == 0 )
13             System.out.println(n);
14         else{
15             for (int i = n; i >=1; i--)
16                 fact *= i;
17                 n--;
18         }
19
20         System.out.println("Factorial of " + " n " + " is " + fact );
21     }
22 }
```

Output:

```
PS D:\Logic Building\Lab\Day 01> cd "d:\Logic
Enter the value of n: 5
Factorial of n is 120
PS D:\Logic Building\Lab\Day 01> 
```

Question 4: Print Even Numbers from 1 to N

Program:

```
J Even.java > ...
1  import java.util.Scanner;
2
3  public class Even {
    Run | Debug
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6          System.out.print("Enter the number: ");
7          int n = sc.nextInt();
8          sc.close();
9
10         for ( int i = 1; i<=n; i++){
11             if (i % 2 ==0)
12                 System.out.print(i + " ");
13         }
14     }
15 }
```

Output:

```
PS D:\Logic Building\Lab\Day 01> cd "d:\Logic Building\Lab\Day 01\" ;
if ($?) { javac Even.java } ; if ($?) { java Even }
Enter the number: 10
2 4 6 8 10
PS D:\Logic Building\Lab\Day 01> 
```

Question 5: Sum of Odd Numbers between 1 and N

Program:

```
J SumOfOdd.java > SumOfOdd
1  import java.util.Scanner;
2
3  public class SumOfOdd {
4      Run | Debug
5      public static void main(String[] args) {
6          Scanner sc = new Scanner(System.in);
7          System.out.print("Enter a number: ");
8          int n = sc.nextInt();
9          sc.close();
10
11          int sum = 0;
12
13          for (int i=0; i<=n; i++){
14              if (i % 2 !=0)
15                  sum += i;
16          }
17          System.out.println("The sum of odd numbers from 1 to " + n + " is " + sum);
18      }
19  }
```

Output:

```
PS D:\Logic Building\Lab\Day 01> cd "d:\Logic Building\Lab\
Enter a number: 10
The sum of odd numbers from 1 to 10 is 25
PS D:\Logic Building\Lab\Day 01> 
```

Question 6: Print All Elements of an Array

Program:

```
J CreateArray.java > CreateArray
1  import java.util.Scanner;
2
3  public class CreateArray {
    Run | Debug
4      public static void main(String[] args) {
5          Scanner sc= new Scanner (System.in);
6
7          int num[] = new int[5];
8
9          System.out.println("Enter 5 integers: ");
10
11         for(int i=0; i < num.length; i++){
12             System.out.print("Enter num[" + i + "] ");
13             num[i] = sc.nextInt();
14         }
15         sc.close();
16
17         System.out.println("The elements of the array: ");
18         for (int n:num)
19             System.out.print(n + " ");
20     }
21 }
```

Output:

```
PS D:\Logic Building\Lab\Day 01> cd
Enter 5 integers:
Enter num[0] 3
Enter num[1] 7
Enter num[2] 12
Enter num[3] 5
Enter num[4] 8
The elements of the array:
3 7 12 5 8
```

Question 7: Find the Sum of All Elements in an Array

Program:

```
import java.util.Scanner;

public class SumOfElements{
    Run | Debug
    public static void main(String[] args) {
        Scanner sc= new Scanner (System.in);

        int num[] = new int[5];

        System.out.println("Enter 5 integers: ");
        for(int i=0; i < num.length; i++){
            num[i] = sc.nextInt();
        }
        sc.close();

        int sum = 0;
        System.out.print("The sum of elements of the array: ");
        for (int n:num)
            sum += n;
        System.out.print(sum);
    }
}
```

Output:

```
PS D:\Logic Building\Lab\Day 01> cd "d:\Logic
Enter 5 integers:
4 6 8 2 10
The sum of elements of the array: 30
PS D:\Logic Building\Lab\Day 01> 
```

Question 8: Print All Names in a Array

Program:

```
ArrayOfNames.java > ArrayOfNames > main(String[])
1  import java.util.Scanner;
2
3  public class ArrayOfNames {
    Run | Debug
4      public static void main(String[] args) {
5          Scanner sc= new Scanner (System.in);
6
7          String names[] = new String[4];
8          System.out.println("Enter 4 names: ");
9
10         for(int i=0; i < names.length; i++){
11             names[i] = sc.next();
12         }
13
14         System.out.println("\nNames in the array: ");
15         for (String n : names)
16             System.out.println(n + " ");
17
18         sc.close();
19     }
20 }
```

Output:

```
PS D:\Logic Building\Lab\Day 01> cd "d:\Logic
Enter 4 names:
John Rosie Mary William

Names in the array:
John
Rosie
Mary
William
PS D:\Logic Building\Lab\Day 01> 
```


Question 9: Find the Largest Element in an Array

Program:

```
J LargestElement.java > ...
3
4 public class LargestElement {
    Run | Debug
5     public static void main(String[] args) {
6         Scanner sc= new Scanner (System.in);
7
8         int num[] = new int[5];
9
10        System.out.println("Enter 5 integers: ");
11
12        for(int i=0; i < num.length; i++){
13            System.out.print("Enter num[" + i + "] ");
14            num[i] = sc.nextInt();
15        }
16        sc.close();
17
18        int largest = Arrays.stream(num).max().orElseThrow();
19        System.out.println("The largest element is: " + largest);
20    }
21 }
```

Output:

```
PS D:\Logic Building\Lab\Day 01> cd "d:\Logic
Enter 5 integers:
Enter num[0] 100
Enter num[1] 250
Enter num[2] 450
Enter num[3] 200
Enter num[4] 800
The largest element is: 800
PS D:\Logic Building\Lab\Day 01> 
```

Question 10: Find the Average of Elements in an Array

Program:

```
J AvgOfElements.java > ...
1  import java.util.Arrays;
2  import java.util.Scanner;
3
4  public class AvgOfElements {
    Run | Debug
5      public static void main(String[] args) {
6          Scanner sc= new Scanner (System.in);
7
8          int num[] = new int[5];
9
10         System.out.println("Enter 5 integers: ");
11         for(int i=0; i < num.length; i++){
12             num[i] = sc.nextInt();
13         }
14         sc.close();
15         double average = Arrays.stream(num).average().orElse(0.0);
16         System.out.println("The average of the elements is: " + average);
17     }
18 }
```

Output:

```
PS D:\Logic Building\Lab\Day 01> cd "d:\Logic
Enter 5 integers:
10 20 70 60 40
The average of the elements is: 40.0
PS D:\Logic Building\Lab\Day 01> 
```

Question 11: Count Positive and Negative Numbers in an Array

Program:

```
J CountPosNeg.java > ...
1  import java.util.Scanner;
2
3  public class CountPosNeg {
    Run | Debug
4      public static void main(String[] args) {
5          Scanner sc= new Scanner (System.in);
6
7          int num[] = new int[6];
8          System.out.println("Enter 6 integers: ");
9          for(int i=0; i < num.length; i++){
10             num[i] = sc.nextInt();
11         }
12         sc.close();
13
14         int pos = 0, neg = 0, zero = 0;
15         for(int i=0; i < num.length; i++){
16             if (num[i] == 0)
17                 zero += 1;
18             else if (num[i] > 0)
19                 pos += 1;
20             else
21                 neg += 1;
22         }
23
24         System.out.println("Elements which are zero: " + zero);
25         System.out.println("Positive elements: " + pos);
26         System.out.println("Negative elements: " + neg);
27     }
28 }
```

Output:

```
PS D:\Logic Building\Lab\Day 01> cd "d:\Logic
Enter 6 integers:
4 -2 6 -3 0 8
Elements which are zero: 1
Positive elements: 3
Negative elements: 2
PS D:\Logic Building\Lab\Day 01> |
```

Question 12: Sort an Array in Ascending Order

Program:

```
J SortAsc.java > SortAsc > main(String[])
1  import java.util.Arrays;
2  import java.util.Scanner;
3
4  public class SortAsc {
    Run | Debug
5      public static void main(String[] args) {
6          Scanner sc= new Scanner (System.in);
7
8          int num[] = new int[5];
9
10         System.out.println("Enter 5 integers: ");
11         for(int i=0; i < num.length; i++)
12             num[i] = sc.nextInt();
13         sc.close();
14
15         Arrays.sort(num);
16         System.out.println("Sorted Array: ");
17         for (int n : num)
18             System.out.print(n + " ");
19     }
20 }
```

Output:

```
PS D:\Logic Building\Lab\Day 01> cd "d:\Logic
Enter 5 integers:
2 11 45 22 80
Sorted Array:
2 11 22 45 80
PS D:\Logic Building\Lab\Day 01> 
```

Question 13: Check if an Array Contains a Specific Element

Program:

```
J Search.java > ...
1  import java.util.Arrays;
2  import java.util.Scanner;
3
4  public class Search {
    Run | Debug
5      public static void main(String[] args) {
6          Scanner sc= new Scanner (System.in);
7
8          Integer num[] = new Integer[5];
9
10         System.out.println("Enter 5 integers: ");
11         for(int i=0; i < num.length; i++)
12             num[i] = sc.nextInt();
13
14         System.out.print("\nEnter a number to search: ");
15         int target = sc.nextInt();
16         sc.close();
17
18         boolean searched = Arrays.asList(num).contains(target);
19
20         if (searched)
21             System.out.println("Found");
22         else
23             System.out.println("Not found");
24     }
25 }
```

Output:

```
PS D:\Logic Building\Lab\Day 01> cd "d:\Logic
Enter 5 integers:
10 20 30 40 50

Enter a number to search: 50
Found
PS D:\Logic Building\Lab\Day 01> |
```

Question 14: Find the Index of an Element in an Array

Program:

```
J Index.java > ...
1  import java.util.Arrays;
2  import java.util.Scanner;
3
4  public class Index {
    Run | Debug
5      public static void main(String[] args) {
6          Scanner sc= new Scanner (System.in);
7
8          Integer num[] = new Integer[5];
9
10         System.out.println("Enter 5 integers: ");
11         for(int i=0; i < num.length; i++)
12             num[i] = sc.nextInt();
13
14         System.out.print("Enter a number to find its index: ");
15         int target = sc.nextInt();
16         sc.close();
17
18         int index = Arrays.asList(num).indexOf(target);
19
20         if (index != -1)
21             System.out.println("The number " + target + " is found at index " + index);
22         else
23             System.out.println("Not found (-1)");
24     }
25 }
```

Output:

```
PS D:\Logic Building\Lab\Day 01> cd "d:\Logic
Enter 5 integers:
10 20 30 40 50
Enter a number to find its index: 50
The number 50 is found at index 4
PS D:\Logic Building\Lab\Day 01> 
```

Question 15: Write a Program to print the following pattern

```
1
2*2
3*3*3
4*4*4*4
5*5*5*5*5
5*5*5*5*5
4*4*4*4
3*3*3
2*2
```

Program:

```
J R_Diamond.java > ...
1  import java.util.Scanner;
2
3  public class R_Diamond {
4      Run | Debug
5      public static void main(String[] args) {
6          Scanner sc = new Scanner(System.in);
7          System.out.print("Enter n: ");
8          int n = sc.nextInt();
9
10         for (int i=0 ; i < n ; i++){
11             for(int j = 0 ; j <= 2*i ; j++){
12                 if (j % 2 == 0)
13                     System.out.print(i+1);
14                 else
15                     System.out.print("*");
16             }
17             System.out.println();
18         }
19
20         for (int i= n ; i >= 2 ; i--){
21             for ( int j = 1 ; j <= 2*i-1 ; j++){
22                 if (j % 2 == 0)
23                     System.out.print("*");
24                 else
25                     System.out.print(i);
26             }
27             System.out.println();
28         }
29         sc.close();
30     }
}
```

Output:

```
Enter n: 5
1
2*2
3*3*3
4*4*4*4
5*5*5*5*5
5*5*5*5*5
4*4*4*4
3*3*3
2*2
PS D:\Logic Building
```


Question 16: Write a Program to print the following pattern

```
1
1*2
1*2*3
1*2*3*4
1*2*3*4*5
```

Program:

```
J Triangle_90.java > Triangle_90 > main(String[])
1  import java.util.Scanner;
2
3  public class Triangle_90 {
4      Run | Debug
5      public static void main(String[] args) {
6          Scanner sc = new Scanner(System.in);
7          System.out.print("Enter n: ");
8          int n = sc.nextInt();
9
10         for (int i=0 ; i < n ; i++){
11             int temp = 1;
12
13             for (int j= 0 ; j <= 2*i ; j++){
14                 if ( j % 2 == 0){
15                     System.out.print(temp++);
16                 }
17                 else
18                     System.out.print("*");
19             }
20             System.out.println();
21         }
22         sc.close();
23     }
```

Output:

```
PS D:\Logic Building\Lab\Day 01> cd "d:\Logic
Enter n: 5
1
1*2
1*2*3
1*2*3*4
1*2*3*4*5
PS D:\Logic Building\Lab\Day 01> 
```

Question 17: Write a Program to print the following pattern

```
1
1*3
1*3*5
1*3*5*7
1*3*5*7*9
```

Program:

```
J T2_90.java > T2_90 > main(String[])
1  import java.util.Scanner;
2
3  public class T2_90 {
    Run | Debug
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6          System.out.print("Enter n: ");
7          int n = sc.nextInt();
8
9          for (int i=0 ; i < n ; i++){
10             int temp = 1;
11             for (int j= 0 ; j <= 2*i ; j++){
12                 if ( j % 2 == 0){
13                     System.out.print(temp);
14                     temp += 2;
15                 }
16                 else
17                     System.out.print("*");
18             }
19             System.out.println();
20         }
21         sc.close();
22     }
23 }
```

Output:

```
PS D:\Logic Building\Lab\Day 01> cd "d:\Logic
Enter n: 5
1
1*3
1*3*5
1*3*5*7
1*3*5*7*9
PS D:\Logic Building\Lab\Day 01> 
```

Question 18: Write a Program to print the following pattern

```
11111
22222
33333
44444
55555
```

Program:

```
J RectOfNum.java > RectOfNum > main(String[])
1  import java.util.Scanner;
2
3  public class RectOfNum {
    Run | Debug
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6          System.out.print("Enter n: ");
7          int n = sc.nextInt();
8
9          for (int i=0 ; i < n ; i++){
10             for (int j = 0 ; j < n ; j++)
11                 System.out.print(i+1);
12             System.out.println();
13         }
14         sc.close();
15     }
16 }
17
```

Output:

```
PS D:\Logic Bui
Enter n: 5
11111
22222
33333
44444
55555
```

Question 19: Write a Program to print the following pattern

```
1
22
333
4444
55555
```

Program:

```
J T3_90.java > T3_90 > main(String[])
1  import java.util.Scanner;
2
3  public class T3_90 {
    Run | Debug
4  public static void main(String[] args) {
5      Scanner sc = new Scanner(System.in);
6      System.out.print("Enter n: ");
7      int n = sc.nextInt();
8
9      for (int i=0 ; i < n ; i++){
10         for (int j= 0 ; j <= i ; j++)
11             System.out.print(i+1);
12
13             System.out.println();
14         }
15         sc.close();
16     }
17 }
18
```

Output:

```
PS D:\Logic Bui
Enter n: 5
1
22
333
4444
55555
PS D:\Logic Bui
```

Question 20: Write a Program to print the following pattern

```
1
12
123
1234
12345
```

Program:

```
J T4_90.java > ...
1  import java.util.Scanner;
2
3  public class T4_90 {
    Run | Debug
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6          System.out.print("Enter n: ");
7          int n = sc.nextInt();
8
9          for (int i=0 ; i < n ; i++){
10             int temp = 1;
11             for (int j= 0 ; j <= i ; j++)
12                 System.out.print(temp++);
13
14             System.out.println();
15         }
16         sc.close();
17     }
18 }
```

Output:

```
PS D:\Logic Buildi
Enter n: 5
1
12
123
1234
12345
PS D:\Logic Buildi
```

Question 21: Write a Program to print the following pattern

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

Program:

```
J T5_90.java > ...
1  import java.util.Scanner;
2
3  public class T5_90 {
    Run | Debug
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6          System.out.print("Enter n: ");
7          int n = sc.nextInt();
8
9          int temp = 1;
10         for (int i=0 ; i < n ; i++){
11             for (int j= 0 ; j <= i ; j++)
12                 System.out.print(temp++ + " ");
13
14             System.out.println();
15         }
16         sc.close();
17     }
18 }
```

Output:

```
PS D:\Logic Buildi
Enter n: 5
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
PS D:\Logic Buildi
```

Question 22: Write a Program to print the following pattern

```
*****
*       *
*       *
*       *
*       *
*       *
*****
```

Program:

```
HollowRect.java > ...
1  import java.util.Scanner;
2
3  public class HollowRect {
    Run | Debug
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6          System.out.print("Enter n: ");
7          int n = sc.nextInt();
8
9          for (int i=0 ; i < n ; i++){
10             for (int j = 0 ; j < n ; j++){
11                 if (i == 0 || i == n-1 || j == 0 || j == n-1 )
12                     System.out.print("*");
13                 else
14                     System.out.print(" ");
15             }
16             System.out.println();
17         }
18         sc.close();
19     }
20 }
```

Output:

```
PS D:\Logic Build> Enter n: 6
*****
*       *
*       *
*       *
*       *
*****
```

Question 23: Write a Program to print the following pattern

Program:

```
1 import java.util.Scanner;
2
3 public class Diamond {
4     Run | Debug
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7         System.out.print("Enter n: ");
8         int n = sc.nextInt();
9
10        for (int i=0 ; i < n ; i++){
11            for (int j = 0 ; j < n-i-1 ; j++){
12                System.out.print(" ");
13            }
14            for (int j = 0 ; j <= 2*i ; j++){
15                System.out.print("*");
16            }
17            System.out.println();
18        }
19        for (int i = 1 ; i < n ; i++){
20            for (int j = 1 ; j <= i ; j++){
21                System.out.print(" ");
22            }
23            for (int j = ((2*n-1)-(2*i)) ; j > 0 ; j--){
24                System.out.print("*");
25            }
26            System.out.println();
27        }
28        sc.close();
29    }
30 }
```


Output:

```
PS D:\Logic Buil  
Enter n: 6  
      *  
    ***  
   *****  
  *********  
 ****  
*****  
*****  
*****  
*****  
*****  
*****  
***  
*
```

PS D:\Logic Buil

Question 24: Reverse a String

Program:

```
POOJA > Java > J ReverseString.java > ReverseString > Rev(String)
1  import java.util.Scanner;
2
3  public class ReverseString{
4      public static String Rev(String s){
5          if (s == null) return null;
6
7          char[] arr = s.toCharArray();
8
9          int left = 0;
10         int right = arr.length - 1;
11         while (left < right) {
12             char temp = arr[left];
13             arr[left] = arr[right];
14             arr[right] = temp;
15
16             left++;
17             right--;
18         }
19         return new String(arr);
20     }
21
22     Run | Debug
23     public static void main(String[] args) {
24         Scanner sc = new Scanner(System.in);
25         System.out.print("Enter a string: ");
26         String str = sc.next();
27         sc.close();
28
29         String reversed = Rev(str);
30         System.out.println("\nReversed string: " + reversed);
31     }
32 }
```

Output:

```
Enter a string: hello
Reversed string: olleh
PS D:\P00JA\Java>
```

Question 25: Count Vowels in a String

Program:

```
J CountVowels.java > ...
1  import java.util.Scanner;
2
3  public class CountVowels {
    Run | Debug
4      public static void main(String[] args) {
5          Scanner sc= new Scanner (System.in);
6          System.out.println("Enter a string: ");
7          String str = sc.nextLine();
8          sc.close();
9
10         int count = 0;
11         str = str.toLowerCase();
12
13         for (int i = 0; i < str.length(); i++){
14             char ch = str.charAt(i);
15
16             if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u')
17                 count++;
18         }
19         System.out.println("Total number of vowels: " + count);
20     }
21 }
```

Output:

```
PS D:\Logic Building\Lab\Day 01> cd "d:\Logic
Enter a string:
Elegance
Total number of vowels: 4
PS D:\Logic Building\Lab\Day 01>
```

Question 26: Check if a String is a Palindrome

Program:

```
J Palindrome.java > Palindrome > main(String[])
1  import java.util.Scanner;
2
3  public class Palindrome {
    Run | Debug
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6          System.out.print("Enter a string: ");
7          String str = sc.next();
8          sc.close();
9
10         int left = 0;
11         int right = str.length() - 1;
12         boolean isPalindrome = true;
13
14         while (left < right) {
15             if (str.charAt(left) != str.charAt(right)) {
16                 isPalindrome = false;
17                 break;
18             }
19             left++;
20             right--;
21         }
22
23         if (isPalindrome) {
24             System.out.println(str + " is a palindrome");
25         } else {
26             System.out.println(str + " is not a palindrome");
27         }
28     }
29 }
```

Output:

```
PS D:\Logic Building\Lab\Day 01> cd "d:\Logic
Enter a string: madam
madam is a palindrome
PS D:\Logic Building\Lab\Day 01> |
```

Question 27: String Literal and Object Creation

Program:

```
J StringLiterals.java > StringLiterals > main(String[])
1 public class StringLiterals {
    Run | Debug
2     public static void main(String[] args) {
3         String str1 = "hello";
4         String str2 = "hello";
5
6         boolean result = (str1 == str2);
7         System.out.println("Both variables point to the same object: " + result );
8     }
9 }
```

Output:

```
PS D:\Logic Building\Lab\Day 01> cd "d:\Logic Bui
Both variables point to the same object: true
PS D:\Logic Building\Lab\Day 01> 
```

Question 28: String Creation with new Keyword

Program:

```
J StringNew.java > ...
1 public class StringNew {
    Run | Debug
2     public static void main(String[] args) {
3         String str1 = new String("hello");
4         String str2 = new String("hello");
5
6         boolean resultEqualsOperator = (str1 == str2);
7
8         boolean resultEqualsMethod = str1.equals(str2);
9
10        System.out.println("Using == : " + resultEqualsOperator);
11        System.out.println("Using equals() : " + resultEqualsMethod);
12    }
13 }
```

Output:

```
PS D:\Logic Building\Lab\Day 01> cd "d:\Logic
Using == : false
Using equals() : true
PS D:\Logic Building\Lab\Day 01> 
```

Question 29: String Concatenation and Object Creation

Program:

```
Concatenation.java > ...
1  public class Concatenation {
    Run | Debug
2  public static void main(String[] args) {
3      String str1 = "hello";
4      String str2 = "world";
5      String str3 = str1 + str2;
6
7      boolean isSameObject = (str3 == str1);
8
9      System.out.println("Is str3 pointing to the same object as str1? " + isSameObject);
10 }
11 }
```

Output:

```
PS D:\Logic Building\Lab\Day 01> cd "d:\Logic Building\"
Is str3 pointing to the same object as str1? false
PS D:\Logic Building\Lab\Day 01> 
```

Question 30: String Pool with intern() Method

Program:

```
StringIntern.java > ...
1  public class StringIntern {
    Run | Debug
2  public static void main(String[] args) {
3      String str1 = new String("hello");
4      String str2 = str1.intern();
5      String str3 = "hello";
6
7      boolean isSameObject = (str2 == str3);
8
9      System.out.println("Is str2 and str3 pointing to the same object? " + isSameObject);
10 }
11 }
```

Output:

```
PS D:\Logic Building\Lab\Day 01> cd "d:\Logic Building\"
Is str2 and str3 pointing to the same object? true
PS D:\Logic Building\Lab\Day 01> 
```

Question 31: Multiple String Literals with Same Content

Program:

```
J MultipleStringLiterals.java > ...  
1 public class MultipleStringLiterals {  
    Run | Debug  
2     public static void main(String[] args) {  
3  
4         String str1 = "java";  
5         String str2 = "java";  
6         String str3 = "java";  
7  
8         boolean allSame = (str1 == str2) && (str2 == str3);  
9  
10        System.out.println("All strings point to the same object: " + allSame);  
11    }  
12 }
```

Output:

```
PS D:\Logic Building\Lab\Day 01> cd "d:\Logic Bui  
All strings point to the same object: true  
PS D:\Logic Building\Lab\Day 01> 
```